SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
PRACTICAL ASSIGNMENT 11

1. Student Details

Student Name	Homesh Arvind Sawarkar
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Division	A
Batch	A2
Roll No.	202501110069
Date of Submission	27/10/2026

2. Practical Details

Practical No.	11
Practical Title	STANDARD TEMPLATE LIBRARY
Date Performed	-
Faculty Name	Khushal Khairnar

3. Problem Statement

a) Assignment based on list class as a container of standard template library. b) Assignment based on vector class as a container of the standard template library.

4. Objectives

1. To understand the concepts of list and vector containers in the C++ Standard Template Library (STL).
2. To implement insertion, deletion, traversal, and sorting operations using list and vector containers.
3. To compare the features and applications of list and vector containers in managing collections of data.

5. Expected Learning Outcomes

After completing this practical, the student will be able to:

1. Explain the working and features of STL list and vector containers.
2. Implement insertion, deletion, traversal, and sorting operations using both containers.
3. Differentiate between list and vector based on memory organization, element access, and operation efficiency.

6. Theory / Concept

The Standard Template Library (STL) is a part of C++ that provides ready-made containers, algorithms, and iterators to simplify programming.

List Container: The list container, available in the <list> header, generally uses a doubly linked list to store elements. It supports efficient insertion and deletion at different positions when an appropriate iterator is available. However, it does not support direct indexing.

Vector Container: The vector container, available in the <vector> header, stores elements in contiguous memory locations. It can dynamically increase or decrease in size and supports direct indexing. Insertion and deletion in the middle may require shifting elements.

7. Algorithm / Pseudocode

START

Step 1: Include the required header files: <iostream>, <list>, <vector>, and <algorithm>.

Step 2: Create a list and a vector of integers and insert elements into both containers.

Step 3: Display the elements of both containers using loops.

Step 4: Perform insertion, deletion, and sorting operations on both containers using their respective STL functions.

Step 5: Display the updated elements and the size of each container to observe their behavior.

STOP

08. Technology / Tools / Libraries Used

Technology / Tool	**Purpose**
Programming Language	C++
IDE / Platform	VS Code
Libraries / Packages	-

09. Implementation / GitHub Repository

Particular	Details
GitHub Repository URL	https://github.com/Homeshsawarkar/Assignment-11-C-
Repository Name	Assignment 11 C++
Branch Used	main
Commit ID / Latest Commit	_____
Date of Last Commit	Oct 27, 2026

10. Results / Observations



11. Conclusion

In this practical, the STL list and vector containers were implemented in a single C++ program. Insertion, deletion, traversal, sorting, and size operations were performed on both containers. The practical demonstrated their differences in memory organization, element access, and operation efficiency, helping develop an understanding of selecting suitable STL containers for different programming requirements.

12. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

Student Name	PRN	Signature
Homesh Sawarkar	202501110069	Homesh A.S